# Jason Thomas, PhD

Healthcare Data Operations & AI/MLOps Portfolio Strategy Leader | Multimodal Medical Data & Partnerships

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## SUMMARY


- Healthcare data, AI/MLOps, and regulated medical-data platform leader with 14+ years in healthcare and medical research; currently head Data & AI Platform Engineering for Philips Image Guided Therapy Devices R&D.
- Set healthcare data portfolio strategy and MLOps/platform architecture across a ~450-person global medtech R&D organization; currently formalizing a BU-wide R&D Data & AI Platforms Strategy that translates product roadmaps into data-demand/supply models, acquisition/annotation priorities, fit-for-purpose criteria, platform roadmap, governance/QMS controls, investment metrics, and executive dashboards.
- Operate at business-unit senior-leadership level: served on the FY2025 ~25-person management team for a ~5,000-person BU, authored the BU's five-year data and AI strategy narrative, and presented on data/AI to EVP extended leadership spanning 15,000 employees.
- Built and lead a distributed 0-to-8 platform team across data sourcing/curation/annotation, data/ML engineering, infrastructure/platform engineering, MLOps, data stewardship/governance, and real-world data/clinical informatics; manage a ~$3M budget, hold decision rights across licensed-data/vendor/tooling choices, and influence $100M+ additional R&D budget.
- Architect healthcare data and MLOps foundations that ingest, curate, validate, govern, de-identify, version, and serve multi-institution/vendor data for 100+ developers/analysts and product AI/algorithm teams across DICOM-linked imaging, EHR/claims, ECG, audit-log, real-world-evidence, and intraprocedural medical data.
- Bring directly relevant multimodal medical and sensor-data experience spanning DICOM-linked imaging, IVUS/X-ray fluoroscopy co-/tri-registration workflows, ECG/VCG/ECGi, 128-electrode body-surface mapping, intracardiac EGMs, CT/MRI/3D geometry, speech/audio biomarkers, and privacy-preserving synthetic/public data assets.


## CORE COMPETENCIES

**Healthcare Data Portfolio Strategy:** roadmap-driven data demand/supply modeling; labeled/unlabeled data strategy; fitness-for-purpose criteria; modality/clinical-domain prioritization; acquisition and annotation portfolios; partner/data readiness; investment metrics; executive dashboards; release/reuse/public-launch planning

**Healthcare Data Lifecycle & Partnerships:** hospital/customer/vendor acquisition; data-use agreements/SOWs; contracting/pricing; staged pilots; contribution standards; ingestion; curation; active learning; annotation operations; de-identification; provenance; governance; quality controls; access provisioning; serving for training/evaluation

**MLOps, Validation & ML Enablement:** data lakes/lakehouses; ClearML; experiment tracking; lineage/OpenLineage; reproducibility; dataset/model versioning; validation frameworks; data contracts; train/validation/test split governance; gold-label regression evaluation; synthetic data; clinical reader studies; build-vs-integrate decisions

**Multimodal Medical, Sensor & Physical-AI Data:** DICOM-linked imaging; IVUS/X-ray fluoroscopy co-/tri-registration; ECG/VCG/ECGi; 128-electrode body-surface mapping; intracardiac EGMs; CT/MRI/3D torso/heart geometry; speech/audio biomarkers; EHR/claims/audit logs; synthetic/simulation-evaluation data

**Cloud, Data, Compute & NVIDIA Platform Fit:** AWS S3/S3 Tables; Glue Catalog; Athena; Redshift Serverless; Snowflake/Redshift POCs; Lambda; SageMaker; OpenSearch; QuickSight Q; Airflow; PySpark; Parquet; CloudFormation; GitHub Actions; Python; SQL; Slurm-managed GPU/HPC workflows; Holoscan, Isaac, Cosmos, BioNeMo, NeMo Data Designer/Curator, Omniverse alignment; OpenAI Codex, Claude Code, GitHub Copilot

**Regulated Medical AI & Healthcare Data:** HIPAA; GDPR; IRB workflows; EU MDR; EU AI Act; EU Data Act; QMS-aligned controls; de-identification; provenance; governance; DICOM-linked clinical/imaging R&D; real-world evidence; PMCF; OMOP; HL7 FHIR; SNOMED; LOINC; ICD-10; CPT; HCPCS; RxNorm; UMLS

## EXPERIENCE

**Philips Image Guided Therapy Devices - Bothell, WA**

Head of Data & AI Platform Engineering | Oct 2024 - Present; business-unit Data & AI leader since Oct 2024 following leadership transition; role formalized in 2026

Principal Data & AI Engineer | Aug 2024 - Sep 2024

Tech Lead - Senior Data & AI Scientist | Sep 2023 - Jul 2024

Senior Data & AI Scientist | Feb 2022 - Aug 2023

- Own global R&D Data & AI Platforms Strategy for a ~450-person R&D organization, translating product roadmaps into data-demand/supply models, labeled/unlabeled requirements, acquisition and annotation priorities, fit-for-purpose criteria, minimum-viable-to-future-state platform capabilities, governance/QMS/privacy controls, centralized/federated operating model, 5-/10-year roadmap, investment logic, success metrics, and executive dashboard.
- Drive end-to-end healthcare data collaborations across hospitals, Philips customers, vendors, and cross-BU partners: data access paths, data-use agreements/SOWs, licensing decisions, contribution standards, annotation/vendor selection, pricing and staged-pilot paths, release/reuse planning, and conversion of governed clinical/imaging/ex vivo/animal assets into AI, validation, and real-world-evidence data products.
- Prioritize acquisition and platform investments by R&D/model needs, clinical-domain fit, downstream model value, partner/data readiness, cost, contract path, technical feasibility, annotation needs, regulatory constraints, and reuse potential; hold decision rights across licensed data, annotation vendors, data cataloging, warehouse/BI, and lakehouse architecture choices.
- Lead Data & AI Platform Engineering across five pods covering data sourcing/curation/annotation, data/ML engineering, MLOps, data infrastructure/platform engineering, governance, and real-world data/clinical informatics; manage a ~$3M budget and distributed team of 8 with additional dotted-line reports; built the function from 0 to 8, mentored two scientists to tech-lead promotion, and sustained 88% retention through budget freezes.
- Own the clinical data/AI platform ingesting ~80% of R&D data and supporting 100+ developers/analysts through governed self-service access, audit logging, de-identification/privacy controls, access provisioning, data observability with freshness/volume/cost alerts, and data serving for AI development, training/evaluation, analytics, and regulated decision support.
- Architect lakehouse modernization for DICOM-linked clinical/imaging and real-world-evidence data on AWS S3/S3 Tables, Glue Catalog, Athena, bronze/silver/gold data levels, data contracts, cataloging, lineage/OpenLineage, reproducibility controls, ABAC/RBAC/SSO, and ClearML-enabled MLOps patterns; ran hands-on terabyte-scale Snowflake vs. Redshift Serverless POCs and make build-vs-integrate decisions across internal and external systems.

- Establish MLOps and validation workflows for regulated AI development, including experiment tracking, lineage, traceability, reproducibility, dataset/model versioning practices, data quality gates, and FDA-grade edge-inference validation support with patient-matched, site-separated train/validation/test splits and blinded held-out evaluation sets.
- Lead platform patterns for regulated intraprocedural multimodal medical/sensor data used in co-/tri-registration and AI-assisted workflows, including IVUS and X-ray fluoroscopy context; align metadata, quality rules, de-identification/privacy controls, lineage, access, and train/evaluation readiness for AI/algorithm teams and physical-AI/sensor-fusion use cases.
- Standardized reusable AI-ready healthcare data assets, including a versioned library of 150+ computable phenotypes/features adopted across business units and used in regulatory submissions and downstream analytics/AI workflows; enabled 20+ reproducible post-market safety surveillance studies/year and reduced dashboard build time from ~2 days to ~10 minutes via API automation.
- Architected audit-ready real-world-evidence analytics and EU MDR post-market surveillance across >1,000 hospitals and 1B+ patient visits, delivering 10x faster insights at ~75% lower cost, >50% warehouse cost reduction, ~$3M/year savings, and computable-phenotype/site-targeting analytics used in management reviews for TAM estimation, reprocessing calculations, business cases, sales targeting, acquisition analysis, and a $200M recall decision.
- Delivered production AI/GenAI systems on AWS, including Bedrock + Claude identity resolution with F1=0.97, CI/CD-gated gold-label regression evaluation using F1/precision/recall, and >90% reduction in compute and human review; also built REALM, a LangChain + Bedrock + Neo4j hybrid graph/vector RAG platform for explainable device-safety insights.
- Co-chair the business-unit Data Governance Board, defining ownership, quality, metadata, privacy, access, lineage, and QMS-aligned decision processes; served as business-unit lead for EU AI Act/EU Data Act readiness in FY2025, co-authored QMS AI process documentation/training for ~100 people, and supported external-facing regulated work through FDA pre-submissions and FDA presentations.

**NLM Biomedical Informatics & Data Science Predoctoral Fellow - University of Washington; UW Medicine | Sep 2017 - Sep 2021**

- Built OMOP-based data warehouse and pipelines; assessed whether real and synthetic EHR/audit-log data were fit for secondary use under privacy-preserving constraints; contributed to the National COVID Cohort Collaborative (N3C) synthetic data validation workstream and analyzed >1.8M SARS-CoV-2 tests for geospatial/temporal epidemiologic utility.
- Developed multimodal ML models combining speech/audio, language, and clinical history to predict dementia status and evaluate clinical text/voice biomarkers; trained GPU-accelerated ML models on UW Slurm-managed high-performance computing (HPC) infrastructure and AWS and co-first-authored a $1.75M NIH R01 submission on EHR data utility and privacy.
- Led privacy-preserving analysis of 65M health-system SSO log actions and clinician metadata, generated and evaluated synthetic log-data utility, then produced policy recommendations for chief information security officer and usability stakeholders.
- Released/contributed reusable public healthcare data artifacts, including a public medical NLP reproducibility package for Resource and Response Type Classification for Consumer Health Question Answering, code/data for replication, and a 2,882-question annotated medical question-type dataset.

**Senior Research Assistant - OHSU Cardiology, Translational Electrophysiology Lab | Apr 2015 - Aug 2017**

- Developed predictive models and annotation workflows across 100k+ electrocardiograms (ECGs); sourced 104,000 ECGs under HIPAA, mined EHR outcomes for publication-ready analysis, recruited 350+ patients across studies, wrote study protocols, and secured ~7 internal review board (IRB) approvals.
- Collected and analyzed clinical electrophysiology and multimodal sensor/imaging data across observational and interventional studies, including ECGs, 128-electrode body-surface ECG/ECGi recordings, device interrogations, intracardiac EGMs during cath-lab procedures, and co-authored work combining CT/MRI/3D geometry and electrode localization for cardiac activation mapping.

**Clinic Associate & Electrocardiogram Technician - ZOOM+Care | Jun 2013 - Mar 2015**
**Research Assistant - University of Oregon Cardiopulmonary & Respiratory Lab | Feb 2012 - Jun 2013**

## EDUCATION

PhD, Biomedical Informatics (Data Science Specialization) - University of Washington, Seattle, WA (completed in under 4 years)
BS, Human Physiology and Biology (Double Major) - University of Oregon, Eugene, OR

## SELECTED RECOGNITION, PUBLIC DATA & PUBLICATIONS

- IGT Leadership Team Recognition for Generative AI RAG app (Nov 2024); $500k cross-department Generative AI grant, 1 of 6 selected (2024).
- Best Business Impact (NLP) for AWS Bedrock identity resolution, Philips Global Data & AI Conference (Sep 2023); Image Guided Therapy Devices Innovator Award Finalist (highest R&D award in IGTD, Oct 2023).
- JAMIA Editor's Choice first-author paper (May 2022); JAMIA Student Editorial Board Member (2019-2021); JAMIA Editorial Board Member (2021-2023).
- Selected presentations: US Food & Drug Administration presentations (team presentation, 2026; Cardio Rounds, Mar 2024; team presentation, Nov 2024); AMIA Annual Symposium LB06 Panel on synthetic data and N3C (Nov 2021).
- Publications: 20+ peer-reviewed biomedical data/AI publications and 1,000+ Google Scholar citations across healthcare data utility, synthetic data, speech/audio biomarkers, ECG/VCG/ECGi, 3D medical geometry, and medical NLP.